6 A small solenoid of area of cross section $1.6 \times 10^{-3}\,m^2$ is placed inside a larger solenoid of area of cross-section $6.4 \times 10^{-3}\,m^2$, as shown in Fig. 6.1.

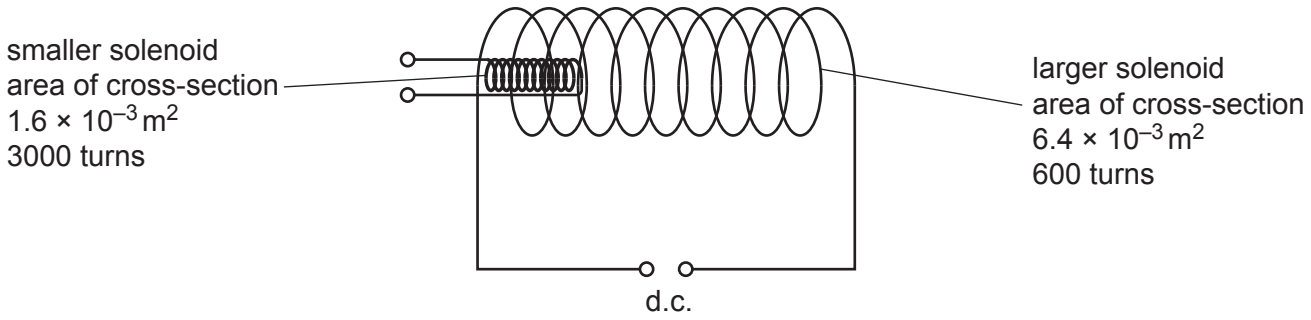


Fig. 6.1 (not to scale)

The larger solenoid has 600 turns and is attached to a d.c. power supply to create a magnetic field.

The smaller solenoid has 3000 turns.

(a) Compare the magnetic flux in the two solenoids.

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.....

..... [1]

(b) Compare the magnetic flux linkage in the two solenoids.

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.....

..... [1]

(c) **(i)** State Lenz's law of electromagnetic induction.

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..... [2]

(ii) The terminals of the smaller solenoid are connected together. The smaller solenoid is then removed from inside the larger solenoid.

With reference to magnetic fields, explain why a force is needed to remove the smaller solenoid.

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..... [3]

[Total: 7]